HEALTH CARE DATA VISUALISATION

Cost Relationship and Policy Insights

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Course	B.Sc CS (AI & DS)
Dataset Used	Healthcare Dataset
Tools Used	Python, Pandas, NumPy, Matplotlib and Seaborn

1.Objective

The objective of this report is to examine patient demographics, billing behaviour, admission patterns, and relationships among age, length of stay, and billing amount. The analysis converts the healthcare dataset into practical insights that can support resource planning, financial monitoring, and future investigation.

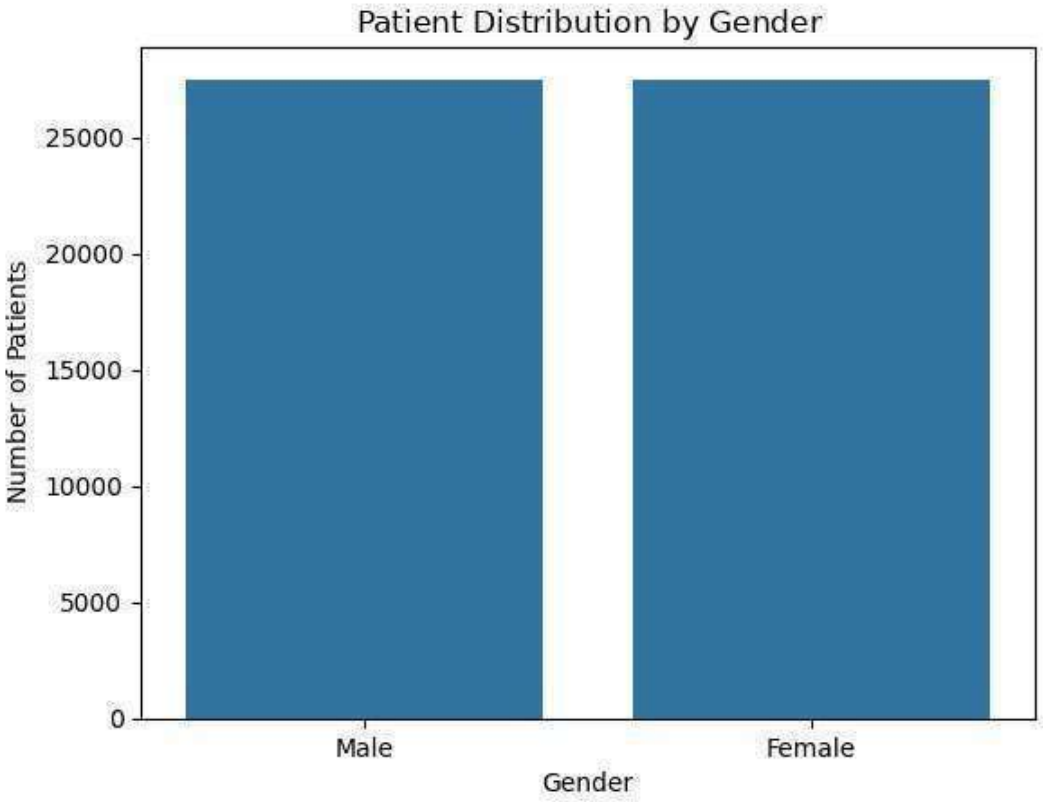
1. Dataset Overview

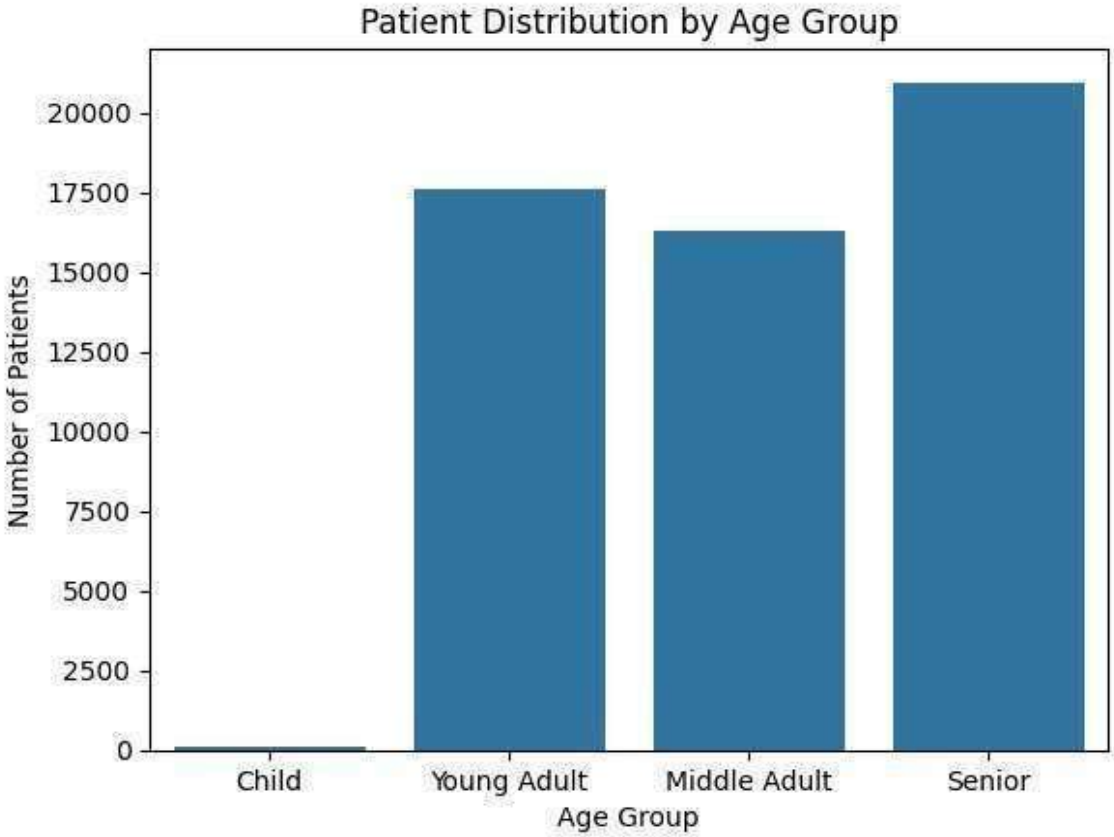
The dataset contains 55,500 patient records and 15 variables, including demographic details, medical condition, admission and discharge dates, admission type, insurance provider, billing amount, medication, and test results. Duplicate-record checking identified 534 duplicates; these records were removed before the visual analysis, leaving 54,966 observations. No missing values were found in the original fields.

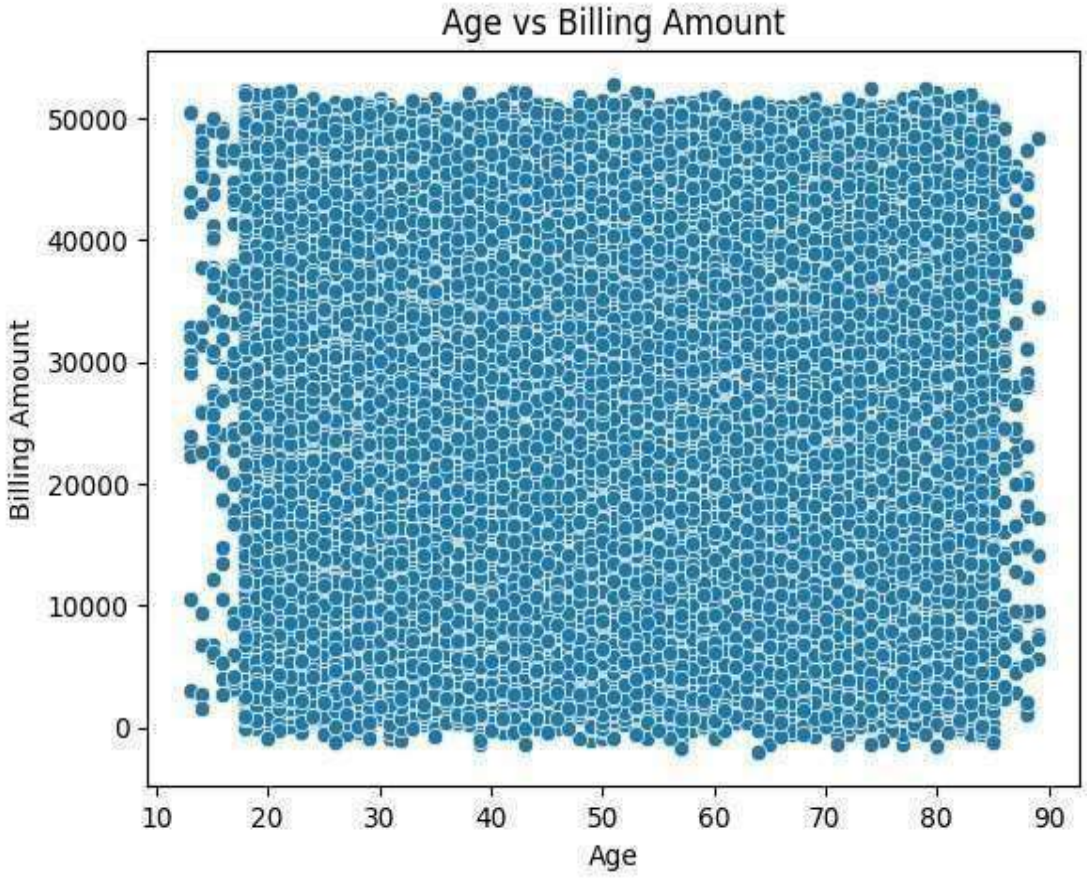
Feature	Value	Interpretation

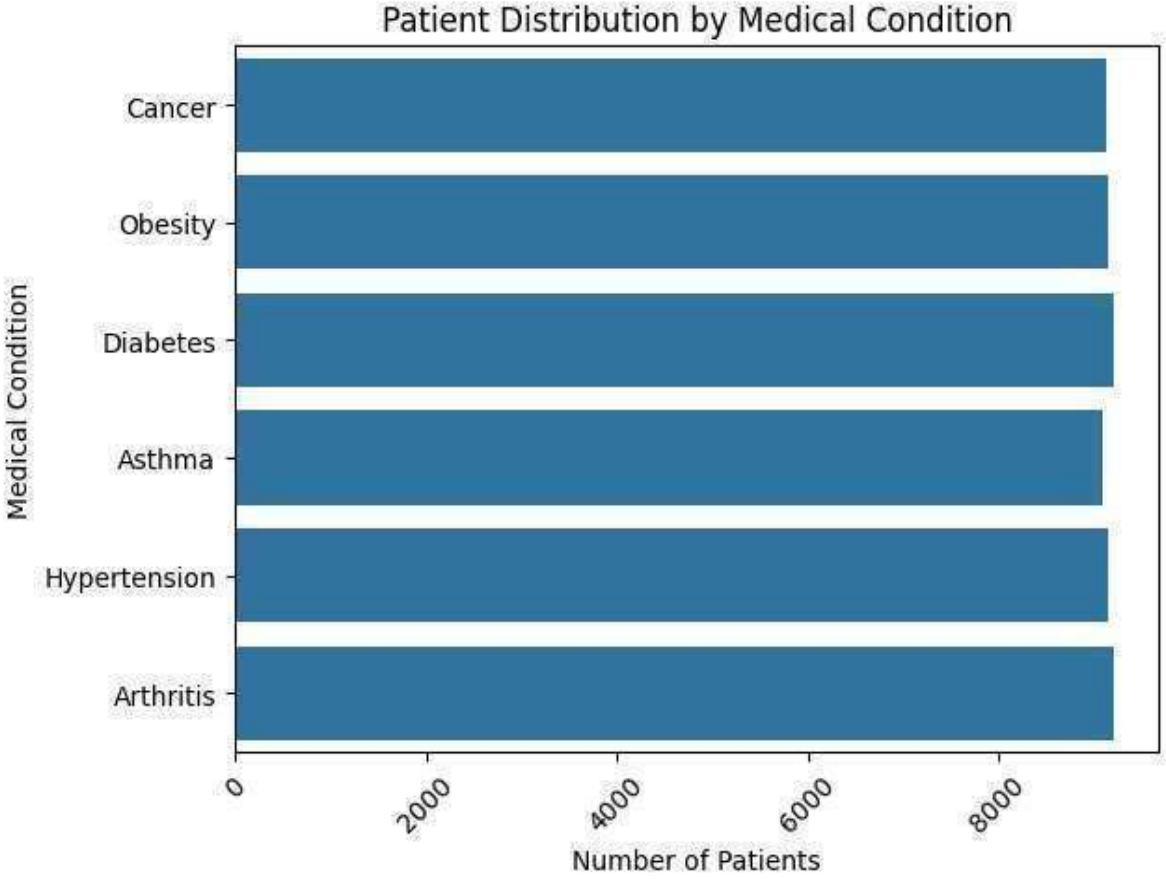
Original records	55,500	Rows available in the source dataset
Variables	15	Demographic, clinical, financial and temporal fields
Duplicate records	534	Removed during preprocessing
Records after cleaning	54,966	Used for all charts and summaries
Conditions	6	Arthritis, Asthma, Cancer, Diabetes, Hypertension and Obesity
Admission types	3	Elective, Urgent and Emergency

2. Patient Demographic Visualisation


Patient Distribution by Gender
Number of Patients
25000
20000
15000
10000
5000
0
Male
Female
Gender


Patient Distribution by Age Group
Number of Patients
20000
17500
15000
12500
10000
7500
5000
2500
0
Child
Young Adult
Middle Adult
Senior
Age Group

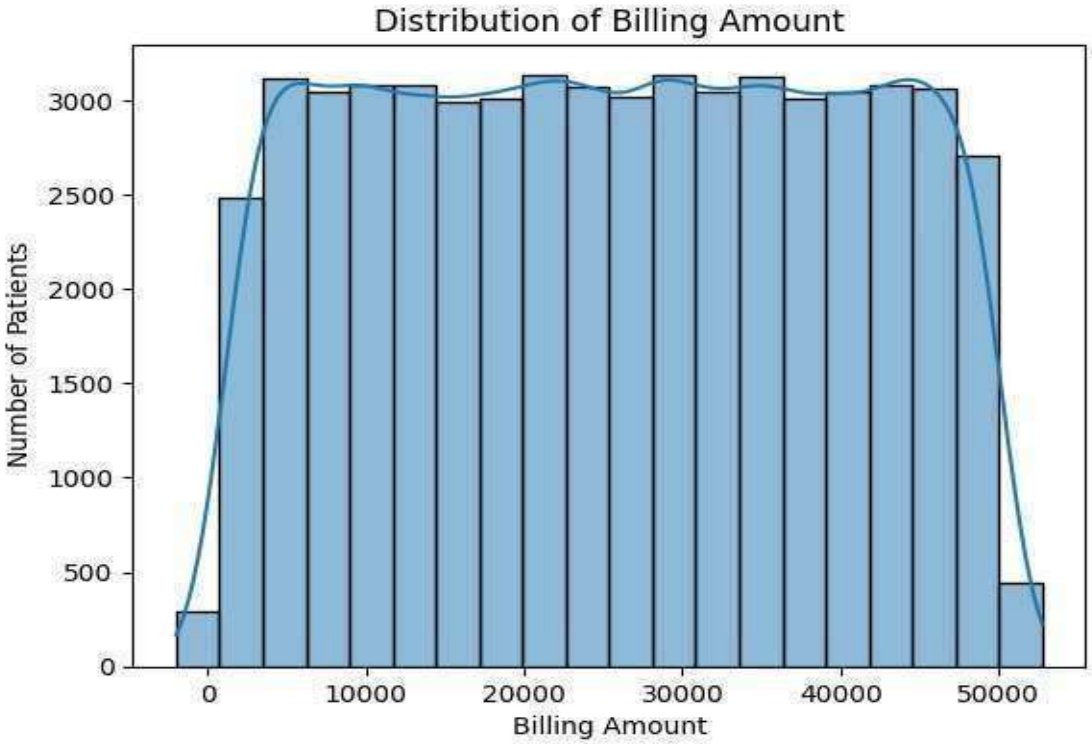

Age vs Billing Amount
Billing Amount
50000
40000
30000
20000
10000
0
10
20
30
40
50
60
70
80
90
Age

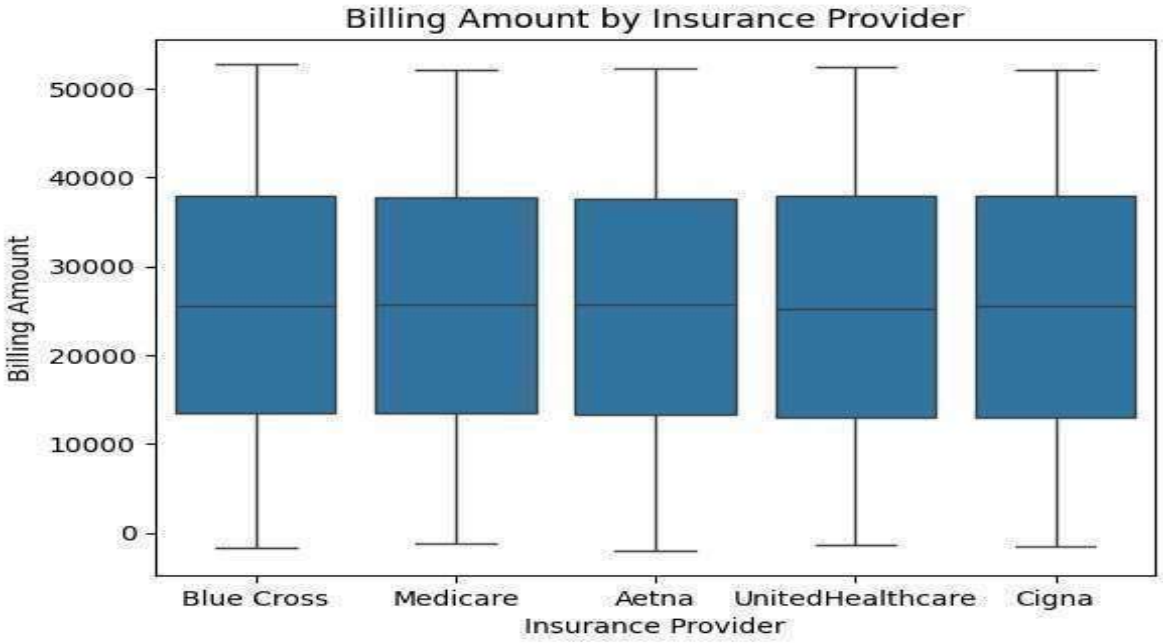

Patient Distribution by Medical Condition
Medical Condition
Cancer
Obesity
Diabetes
Asthma
Hypertension
Arthritis
0
2000
4000
6000
8000
Number of Patients

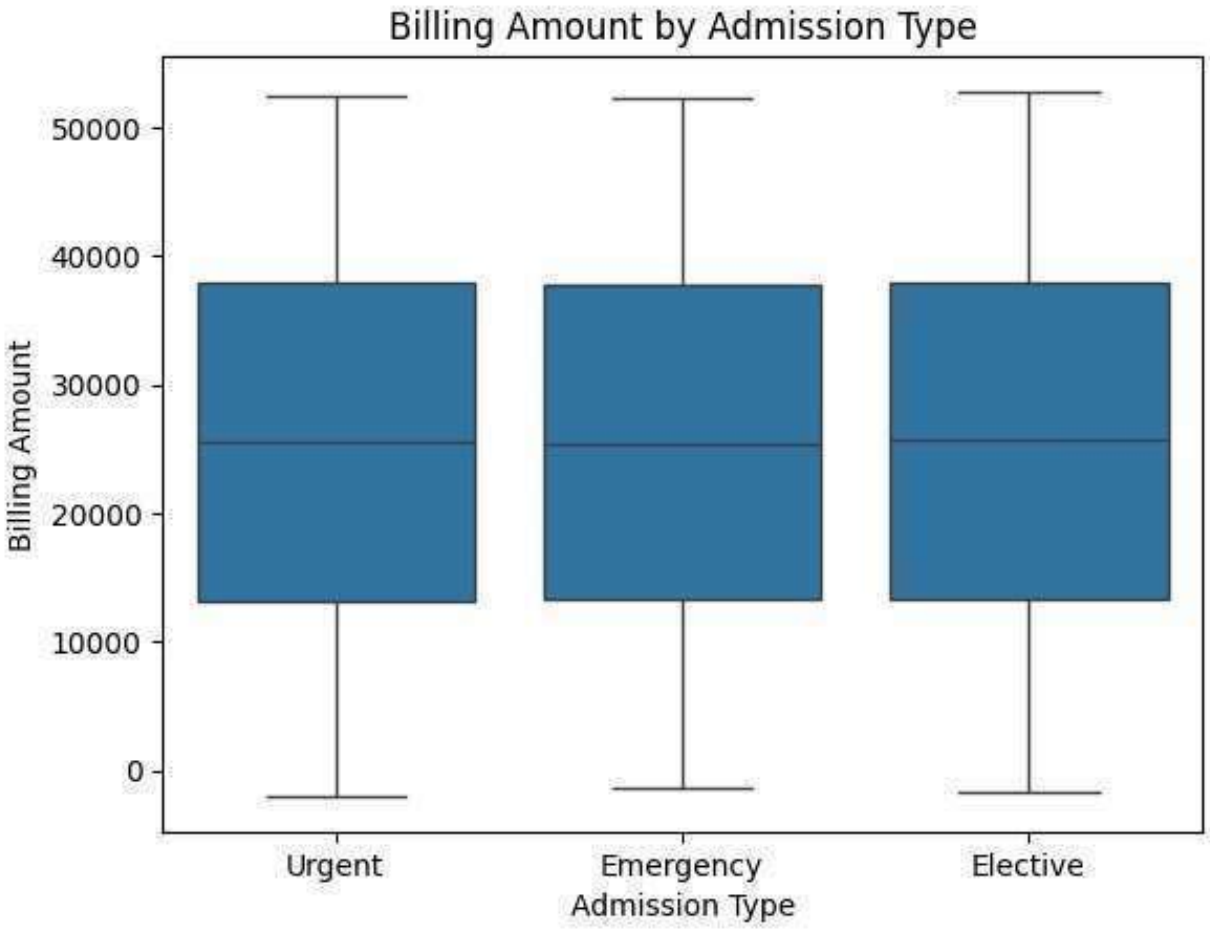
Observation:

Male and female patients are almost equally represented, at roughly 27,500 each. The dataset is dominated by senior patients (20,948), followed by young adults (17,594) and middle adults (16,308). Only 116 records fall into the child category. Medical-condition counts are relatively balanced: Arthritis has 9,218 records, Diabetes 9,216, Hypertension 9,151, Obesity 9,146, Cancer 9,140 and Asthma 9,095. This indicates that staffing and service planning should focus strongly on adult and senior care rather than assuming one condition alone drives the workload.

4.Billing analysis:


Distribution of Billing Amount
Number of Patients
3000
2500
2000
1500
1000
500
0
0
10000
20000
30000
40000
50000
Billing Amount

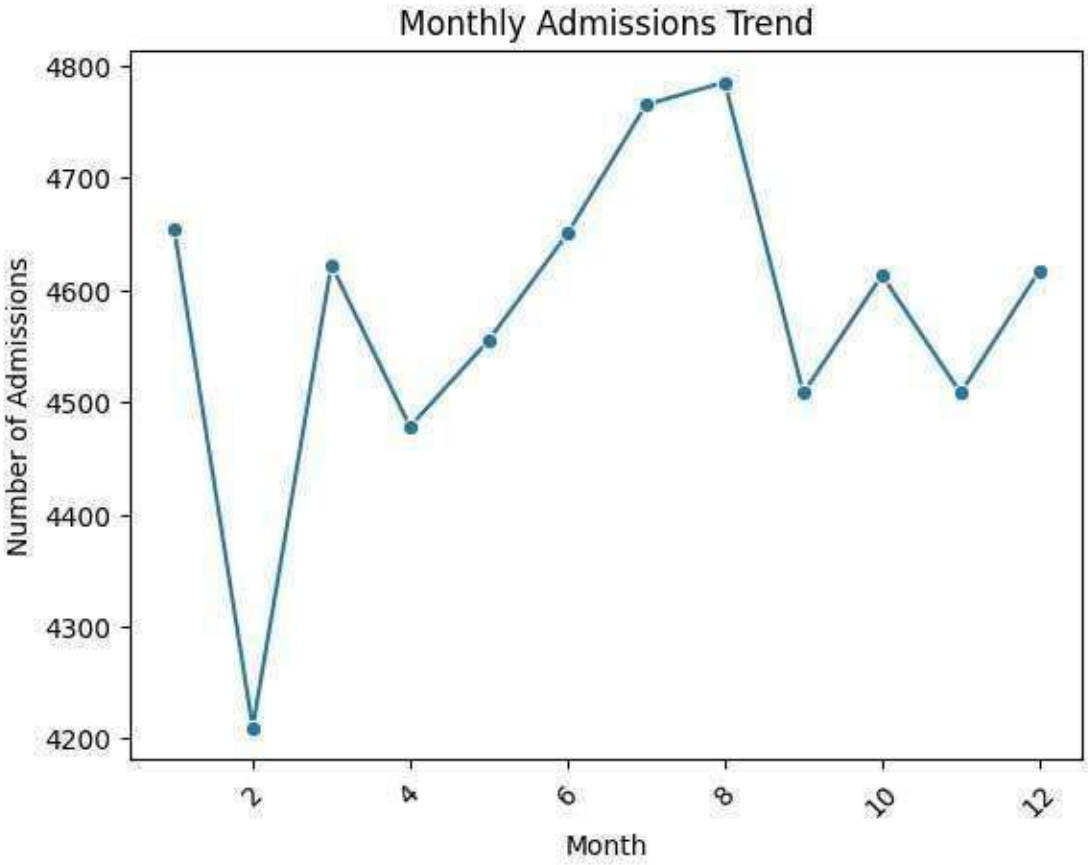

Billing Amount by Insurance Provider
Billing Amount
50000
40000
30000
20000
10000
0
Blue Cross
Medicare
Aetna
UnitedHealthcare
Cigna
Insurance Provider

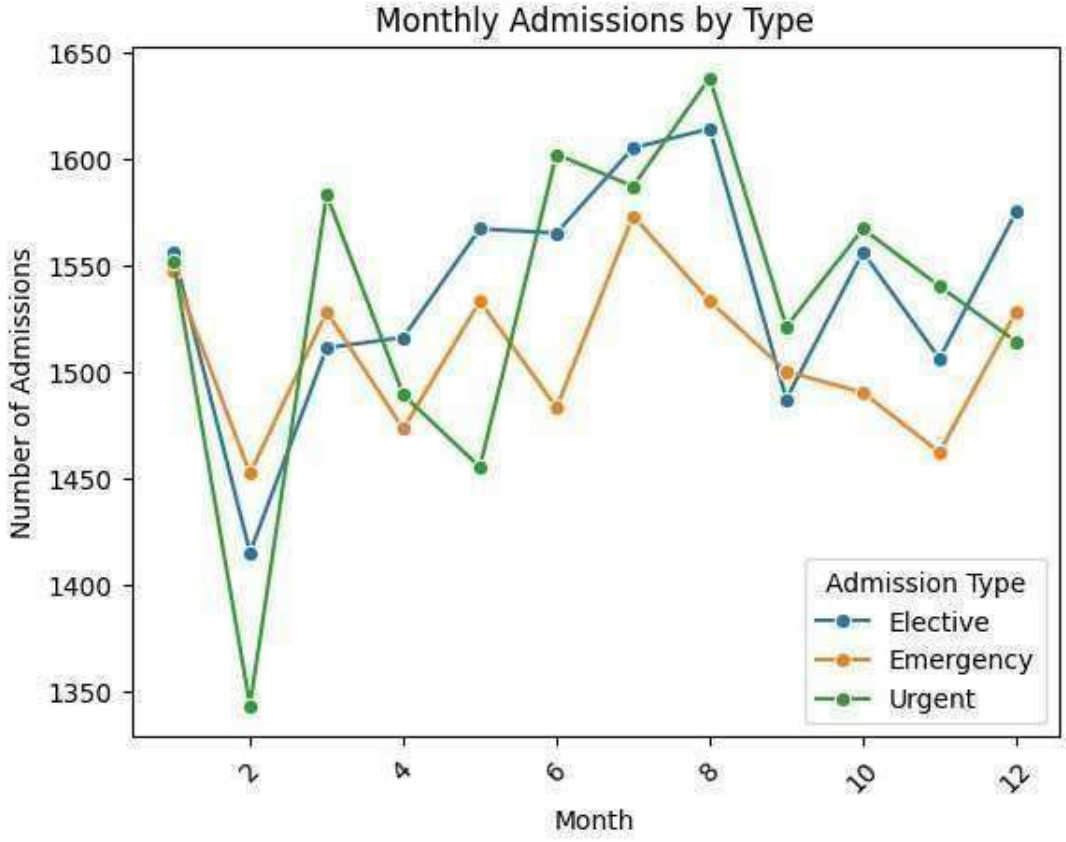


Observation:

The mean billing amount is approximately 25,544.31 after duplicate removal, with a standard deviation of about 14,208.41. The observed range extends from -2,008.49 to 52,764.28, so negative and unusually high values should be audited before financial decisions are made. Insurance-provider and admission-type box plots identify differences in spread and outliers; these comparisons are descriptive and do not prove that a provider or admission type causes higher costs.

5. Temporal Admission Trend Analysis:

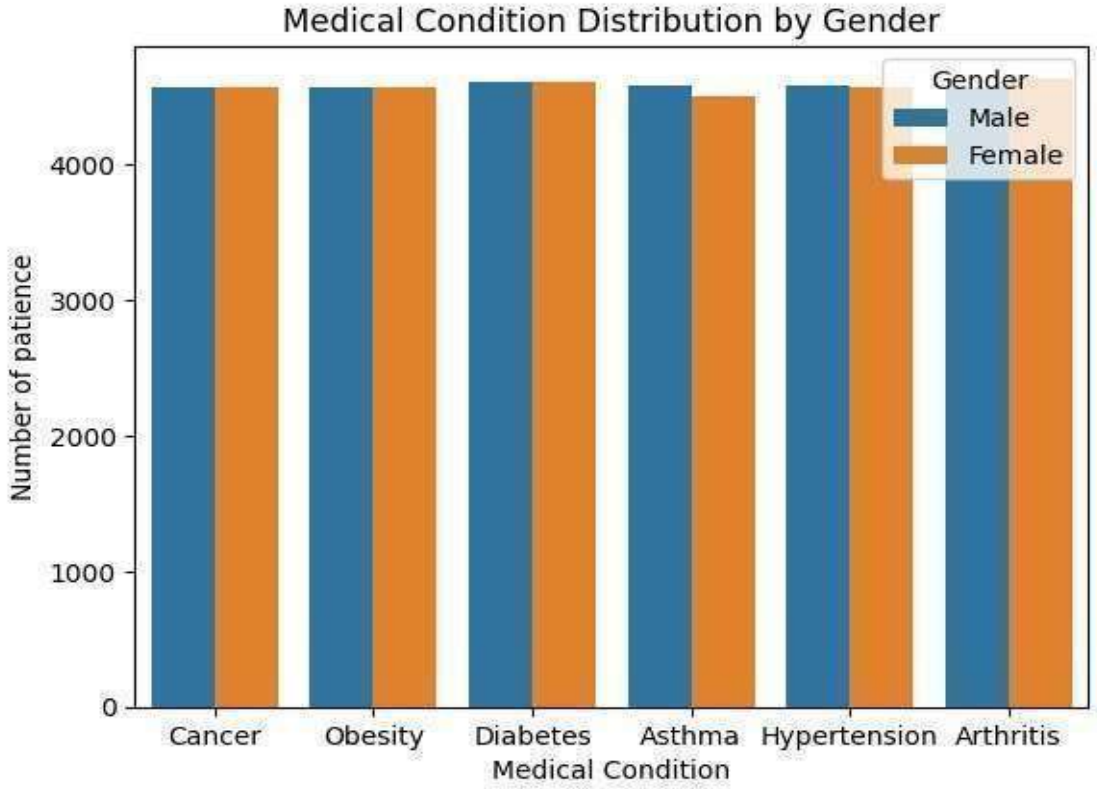

Monthly Admissions Trend
Number of Admissions
4800
4700
4600
4500
4400
4300
4200
2
4
6
8
10
12
Month

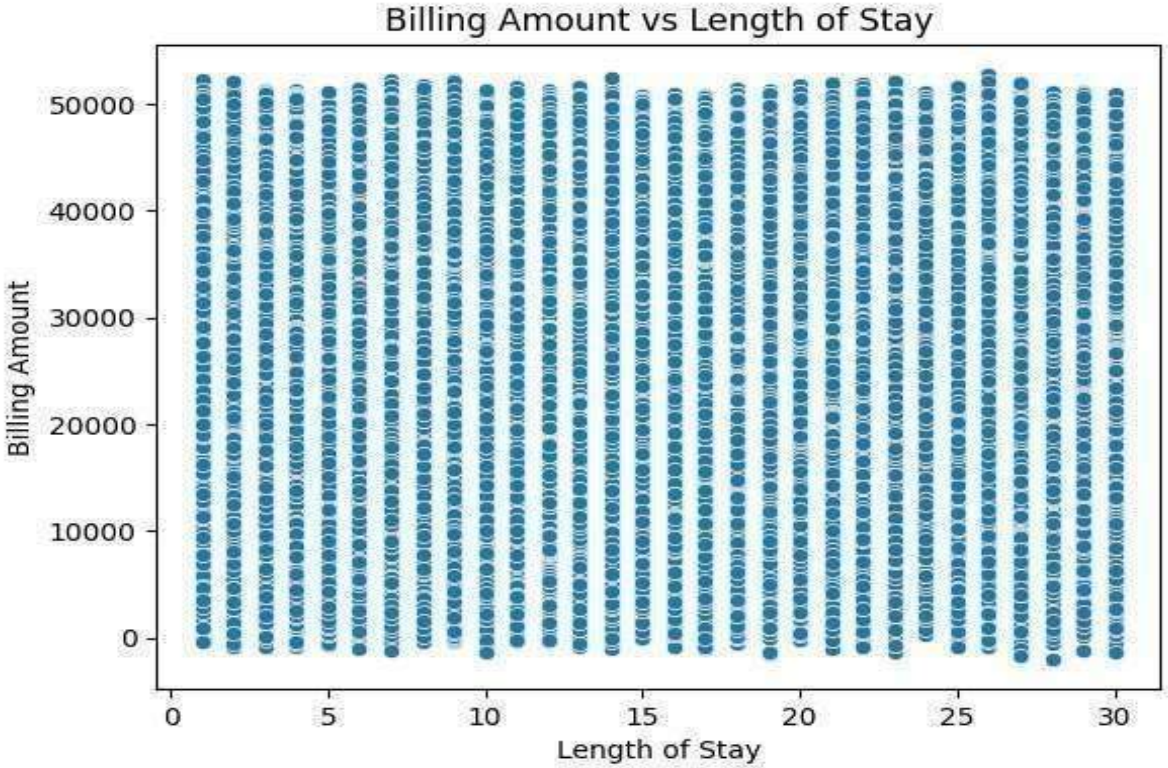

Monthly Admissions by Type
Number of Admissions
1650
1600
1550
1500
1450
1400
1350
2
4
6
8
10
12
Month
Admission Type
Elective
Emergency
Urgent

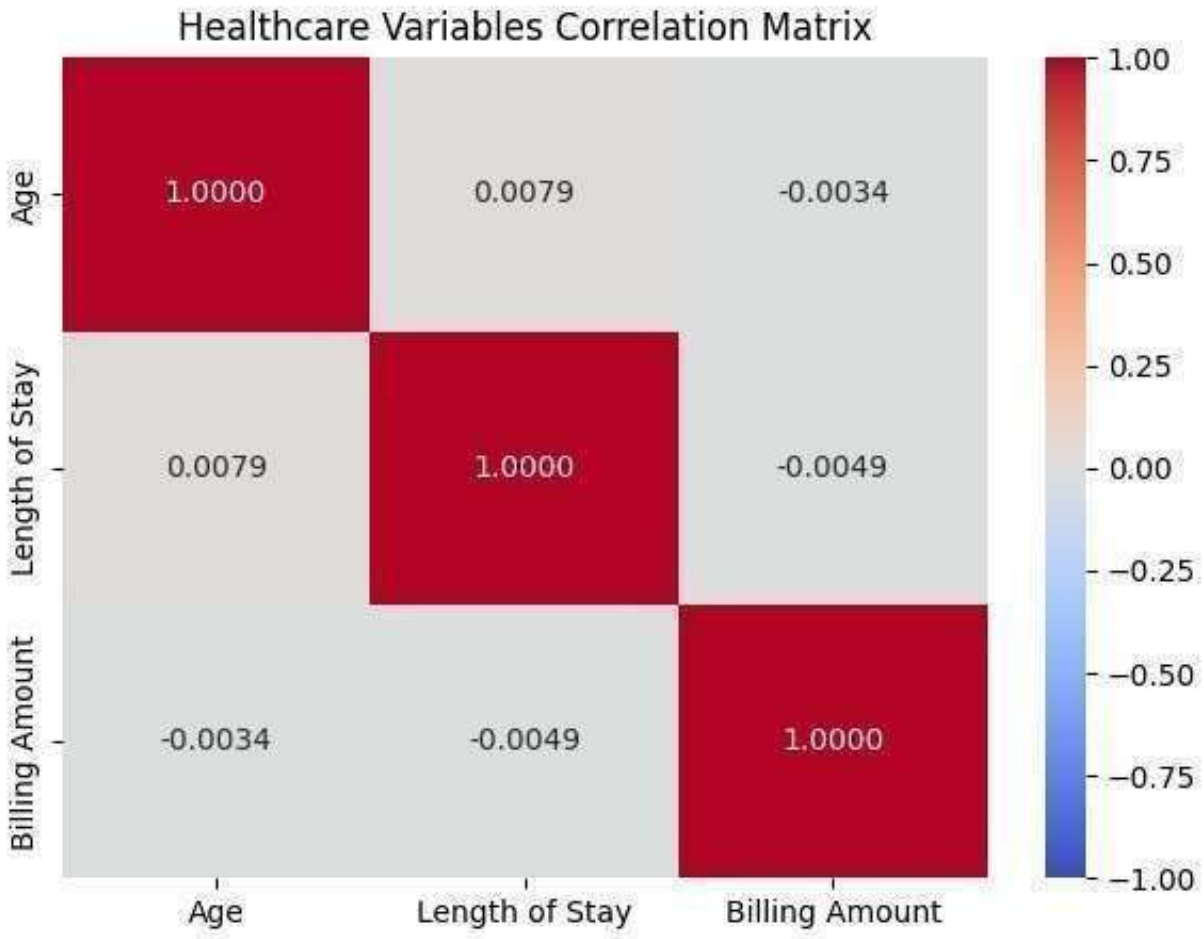
Observation:

Monthly admissions vary from 4,210 in February to 4,785 in August. August is the busiest month in the data, while February is the lowest. The relatively narrow range suggest generally stable annual workload, but the monthly pattern can support rota planning, bed availability, and appointment scheduling.

6. Past Relationship and Correlation Analysis:


Medical Condition Distribution by Gender
Gender
Male
Female
Number of patience
0
1000
2000
3000
4000
Cancer
Obesity
Diabetes
Asthma
Hypertension
Arthritis
Medical Condition


Billing Amount vs Length of Stay
Billing Amount
0
10000
20000
30000
40000
50000
0
5
10
15
20
25
30
Length of Stay



The Pearson correlations are: Age versus Length of Stay = 0.007890; Age versus Billing Amount = -0.003427; Length of Stay versus Billing Amount = -0.004891. These values are all very close to zero, indicating negligible linear association in this dataset. Therefore, age and length of stay alone do not explain billing amount well; other factors such as procedures, treatment intensity, hospital, medication, insurance terms, or data-quality issues should be investigated.

7. Key Findings:

- Senior patients form the largest age group, with 20,948 records, showing the importance of planning services for older adults.
- Medical-condition volumes are nearly balanced, so resources should not be allocated solely around one diagnosis.

- August records the highest monthly admission count at 4,785, while February records the lowest at 4,210.
- Billing amounts show substantial variation, with an approximate mean of 25,544.31 and standard deviation of 14,208.41.
- Age and length of stay have almost zero linear correlation with billing amount, so simple demographic rules are insufficient for cost prediction.

8. Healthcare Management Recommendations:

1. Resource planning:

Use the age-group and monthly-admission results to plan senior-care capacity, staffing, beds and appointment slots. Increase readiness before high-volume periods such as August and review the needs of senior patients separately.

2. Billing analysis:

Create a billing-quality dashboard by insurance provider, admission type, medical condition and hospital. Investigate negative amounts, extreme outliers and inconsistent records before reimbursement or policy decisions.

3. Further investigation:

Extend the analysis with treatment procedures, hospital, doctor, medication, test result and actual resource-use variables. Use multivariable regression or other validated models to determine which factors explain cost and length of stay; correlation alone cannot establish causation.

9. Conclusion:

This analysis provides a structured view of the healthcare dataset through demographic, billing, temporal and correlation visualisations. The strongest operational message is the need to prepare for a predominantly adult and senior patient population while maintaining balanced services across medical conditions. Admissions are broadly stable across months, with a modest peak in August. Billing values vary widely, and the near-zero correlations show that age and length of

stay cannot be used as standalone explanations of cost. Reliable policy decisions therefore require stronger data validation and a broader investigation of clinical and institutional factors.